

The Gas Laws

Answer key with worked solutions

A. Understanding

1. Particles and pressure.

Pressure comes from gas particles colliding with the container walls. In a smaller volume the same particles hit the walls **more often**, so the pressure is higher.

2. Why kelvin?

Volume and pressure are proportional to the **absolute** temperature, which starts at absolute zero (0 K). The Celsius scale starts at an arbitrary point, so ratios of Celsius temperatures are meaningless (and 0 °C would mean dividing by zero).

B. Simple gas laws

3. Boyle.

$$V_2 = \frac{P_1 V_1}{P_2} = \frac{100 \times 5.0}{250} = \mathbf{2.0\ L}$$

4. Charles.

$$T_1 = 293.15\ \text{K}, T_2 = 243.15\ \text{K}. V_2 = 250 \times \frac{243.15}{293.15} = \mathbf{207\ mL}$$

5. Gay-Lussac.

$$T_1 = 288.15\ \text{K}, T_2 = 318.15\ \text{K}. P_2 = 220 \times \frac{318.15}{288.15} = \mathbf{243\ kPa}$$

C. Combined gas law

6. Combined.

$$V_2 = \frac{P_1 V_1 T_2}{P_2 T_1} = \frac{1.00 \times 4.00 \times 373.15}{2.50 \times 298.15} = \mathbf{2.00\ L}$$

D. Ideal gas law

7. Pressure.

$$P = \frac{nRT}{V} = \frac{0.250 \times 0.08206 \times 300}{5.00} = \mathbf{1.23 \text{ atm}}$$

8. Carbon dioxide.

$$\text{(a) } n = \frac{PV}{RT} = \frac{1.50 \times 2.00}{0.08206 \times 300.15} = \mathbf{0.122 \text{ mol}} \quad \text{(b) } M(\text{CO}_2) = 44.01 \text{ g/mol; mass} = 0.12181 \times 44.01 = \mathbf{5.36 \text{ g}}$$

E. Challenge

9. Molar mass.

$$n = \frac{PV}{RT} = \frac{1.00 \times 0.560}{0.08206 \times 273.15} = 0.02498 \text{ mol. } M = \frac{1.00 \text{ g}}{0.02498 \text{ mol}} = \mathbf{40.0 \text{ g/mol}}, \text{ so the gas is argon (39.95 g/mol).}$$